

Netflix Recommender System

SVD Matrix Factorisation vs Item-Based Collaborative Filtering

Netflix Prize Dataset · 3.78M Ratings · 9,980 Users · 4,383 Movies

What Are We Solving?



Core Task

Predict the rating a user would assign to a movie, then surface the Top-10 they have not yet seen.



Key Challenges

Extreme sparsity (91.4%), heavy-tailed popularity, temporal drift, and cold-start.



Dataset

~100M ratings · 480K users · 17,770 movies · Oct 1998–Dec 2005 (Netflix Prize).



Success Metric

Improve RMSE vs baseline (Netflix Prize benchmark), and maximise MAP@10 for ranking quality.

Pipeline & Models

01**load_data.py**

Parse raw Netflix files.
Filter to top 10K active
users → subset CSV.

02**clean_data.py**

Remove dupes, fix
types, filter low-activity
users & rare movies.

03**eda.py**

Generate 8 EDA plots.
Reveal sparsity,
temporal drift,
popularity bias.

04**Temporal Split**

Train: before Oct 2005 /
Test: Oct 2005+
94% / 6% split

05**models.ipynb**

Train SVD & IBCF via
scikit-surprise. Evaluate
RMSE, MAE, MAP@10.

SVD — 50 latent factors · 20 epochs · SGD with L2 regularisation

IBCF — KNN · k=40 · Cosine similarity · Item-based

What the Data Tells Us

Popularity $\neq$ Quality

The Sixth Sense tops volume charts but
Lost: Season 1 leads by average rating.
Models must separate these signals.

Power-Law User Activity

Median user: 334 ratings. Some super-
users exceed 4,000. Bias terms are critical
to avoid skew.

91.4% Sparsity

Even the densest top-user subset is mostly
unrated — latent-factor methods handle
this far better than memory-based CF.

Temporal Drift

Average ratings rose from 2.7 (1999) to
3.5 (2005). A temporal train/test split is
essential; random splits inflate test scores.

TV Box-Set Effect

Season-based titles get disproportionately
high ratings — only dedicated fans rate
them, inflating averages.

Rating Skew

Ratings 3 and 4 dominate (66% of all
ratings). Global mean $\approx$ 3.35 — selection
bias from users only rating content they
chose to watch.

Model Performance

RMSE ↓

MAE ↓

MAP@10 ↑

SVD

0.8885

0.6775

0.7767

Item-Based CF

1.0002

0.7730

0.7014

SVD outperforms IBCF on all three metrics. The MAP@10 gap (0.777 vs 0.701) is especially significant — SVD ranks movies users actually like considerably higher in the top-10 list.

SVD Top-10 — User 2118461

Personalised recommendations for an unseen user — classic family films & beloved TV

1	Inu-Yasha: The Movie 3	5.000	6	Little House on the Prairie S3	4.914
2	Lost: Season 1	5.000	7	The Wizard of Oz: Collector's Ed.	4.902
3	Frosty's Winter Wonderland	5.000	8	Mary Poppins: Bonus Material	4.901
4	Alias: Season 4	4.995	9	Santa Claus Is Comin' to Town	4.896
5	My Fair Lady (Bonus)	4.925	10	An Officer and a Gentleman	4.882

Coherent cluster: classic family films, animated specials, beloved TV series — model correctly identified user taste from sparse ratings.

What's Next

Short-term

SVD++ — add implicit feedback (which movies a user rated, regardless of score); typically 1–3% RMSE gain.

Bias-only baseline — $\mu + b_u + b_i$ is a surprisingly strong benchmark often overlooked.

Hyperparameter grid-search — tune SVD factors (50–200), epochs, and regularisation via CV.

Medium-term

Temporal SVD (timeSVD++) — models time-drifting biases; the method that won the Netflix Prize.

Ensemble blending — stacking SVD + IBCF with a meta-learner typically reduces RMSE by 2–5%.

Long-term

Neural CF / Two-Tower — capture non-linear interactions; incorporate genre, year, and tag features.

LightGCN — graph neural network propagating signals across the user-item bipartite graph.

Context-aware recs — incorporate device, time-of-day, or session context for situational relevance.

Key Takeaways

0.8885

SVD RMSE

0.6775

SVD MAE

0.7767

SVD MAP@10

3.78M

Ratings Analysed

- SVD outperforms IBCF on all metrics. Latent-factor models generalise better over sparse, large-scale rating data.
- Temporal splitting is essential. Random splits inflate test performance by leaking future information into training.
- Popularity bias is real. Volume-based and quality-based signals diverge; models must learn both bias terms explicitly.
- MAP@10 matters more than RMSE alone for real-world recommendation — ranking relevant items to the top is the actual goal.